
Data Analytics Internship

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Domain: Data Analytics

Project: Business Insights Dashboard using Superstore Dataset

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PHASE 2 — BUILD

Project Title

Business Insights Dashboard using Superstore Dataset

Project Type

Individual Project

1. Problem Statement

Businesses generate large amounts of sales data, but it is difficult to extract clear and meaningful insights from it. Traditional reports do not effectively highlight performance trends or loss areas, making it challenging for decision-makers to take informed actions.

2. Introduction

This project focuses on analyzing the Superstore dataset to extract meaningful insights related to sales, profit, and customer behavior. The aim is to understand business performance, identify trends, and support better decision-making using data analysis and visualization techniques.

3. Tech Stack

- Python (Jupyter Notebook) – Data analysis
- Pandas, NumPy – Data processing
- Matplotlib, Seaborn – Data visualization
- Power BI – Dashboard creation
- React.js – Interactive dashboard development

4. Data Cleaning

- Converted the **Order Date** column into proper datetime format for time-based analysis
- Verified dataset structure using shape, columns, and data types
- Checked for missing (null) values to ensure data completeness
- Removed duplicate records to maintain data consistency
- Prepared the dataset for further analysis and visualization

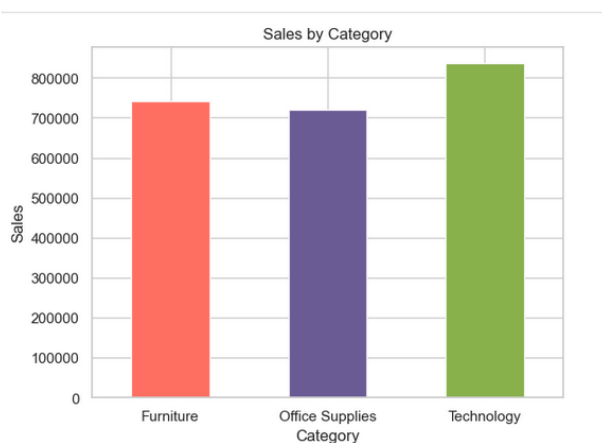
5. Exploratory Data Analysis (EDA)

- Analyzed overall sales and profit distribution
- Compared performance across categories and regions
- Studied time-based trends in sales
- Identified key products and customers contributing to revenue

6. Core Analysis & Visualizations

The following visualizations were created using Jupyter Notebook:

1.Sales by Category (Bar Chart)

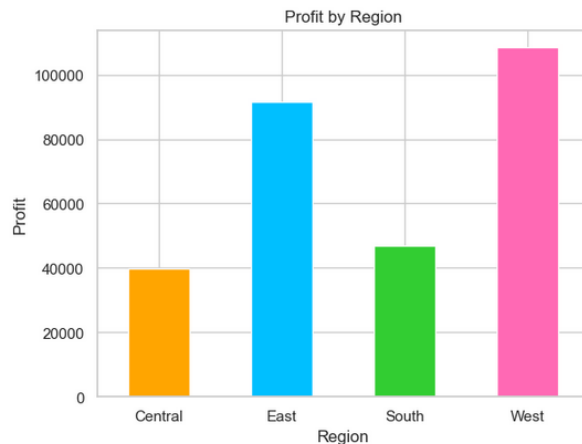


This bar chart represents the total sales generated across different product categories: Furniture, Office Supplies, and Technology. It helps in comparing the contribution of each category to overall revenue.

Insight:

Technology is the top-performing category in terms of sales, suggesting higher demand or higher-value products. Office Supplies shows the lowest sales, indicating a potential area for improvement or lower pricing impact.

2.Profit by Region (Bar Chart)

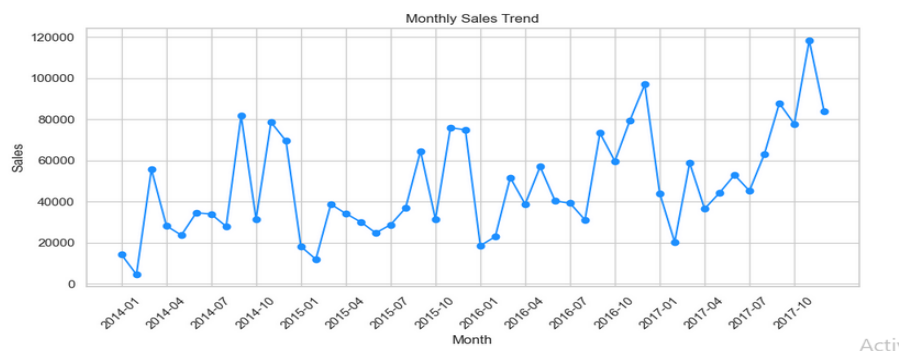


This bar chart shows the total profit generated across different regions: Central, East, South, and West. It helps in comparing the profitability of each region.

Insight

The **West region generates the highest profit**, followed by the East region. The Central region has the lowest profit, indicating weaker performance compared to other regions.

3.Monthly Sales Trend (Line Chart)

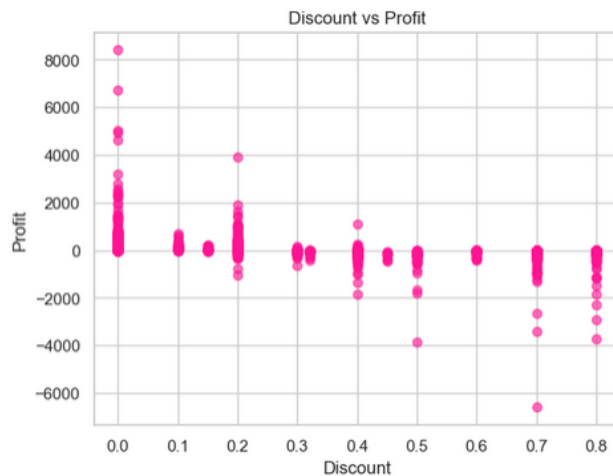


This line chart shows the variation of sales over time on a monthly basis. It helps in identifying trends, patterns, and fluctuations in sales across different months.

Insight

Sales show significant fluctuations over time, with noticeable peaks and drops. The highest sales are observed towards the later months, indicating a possible growth trend or seasonal demand.

4. Discount vs Profit (Scatter Plot)



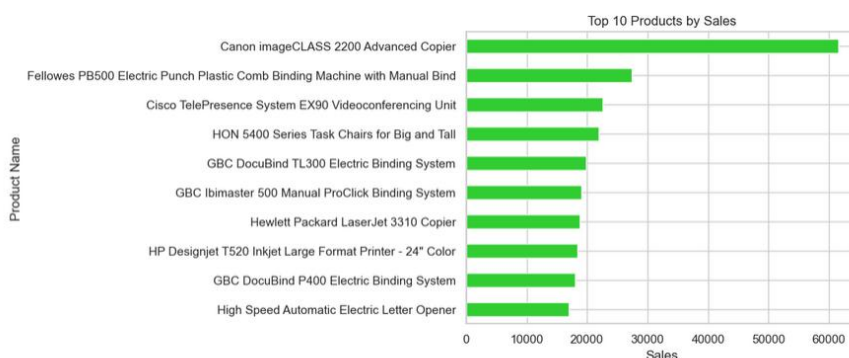
This scatter plot shows the relationship between discount and profit. Each point represents a transaction, helping to understand how different discount levels impact profitability.

Insight

As the discount increases, **profit tends to decrease**, and many transactions with high discounts result in negative profit (loss). This indicates that excessive discounting negatively affects business profitability.

5.Top Products by Sales (Bar Chart)

This horizontal bar chart displays the top 10 products based on total sales. It helps identify which products contribute the most to overall revenue.

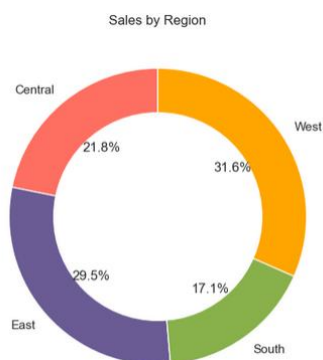


This horizontal bar chart displays the top 10 products based on total sales. It helps identify which products contribute the most to overall revenue.

Insight

The **Canon imageCLASS 2200 Advanced Copier** generates the highest sales among all products, significantly outperforming others. This indicates that a few high-performing products contribute a major share of total revenue.

6. Sales Distribution by Region (Donut Chart)



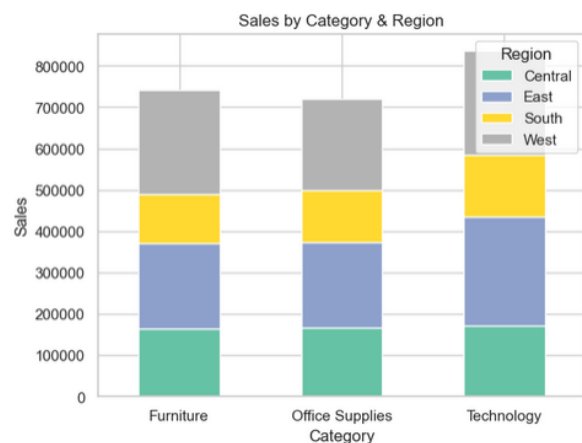
This donut chart represents the percentage contribution of total sales from different regions:

West, East, Central, and South. It helps in understanding which region contributes the most to overall revenue.

Insight

The West region contributes the highest share of sales (31.6%), followed by the East region. The South region has the lowest contribution, indicating comparatively lower sales performance.

7.Sales by Category and Region(Stacked Bar chart)

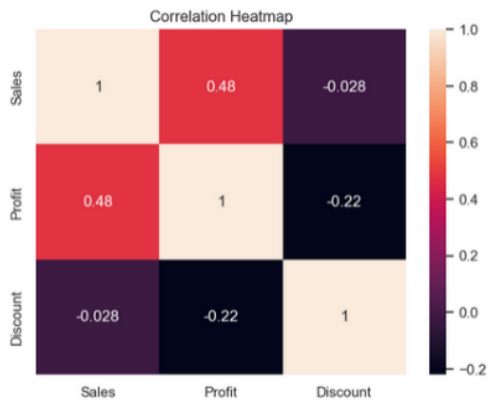


This stacked bar chart shows the distribution of sales across different categories (Furniture, Office Supplies, Technology) and regions (Central, East, South, West). It helps in comparing both category-wise and region-wise performance together.

Insight

The **Technology category has the highest sales across all regions**, with strong contributions from the West and East regions. Overall, the West region consistently contributes a significant portion of sales in each category.

8. Correlation Heatmap



This heatmap shows the correlation between key variables: Sales, Profit, and Discount. It helps in understanding the relationship and dependency between these variables.

Insight

Sales and Profit have a **moderate positive correlation (0.48)**, indicating that higher sales generally lead to higher profit. However, Discount has a **negative correlation with Profit (-0.22)**, showing that increasing discounts reduces profitability.

7. Key Insights

- Technology category generates the highest sales
- Higher discounts negatively impact profitability
- Profit varies across different regions
- A small number of products contribute a major portion of revenue
- Sales fluctuate over time, indicating seasonal trends

8. Statistical Validation

Correlation analysis between Sales, Profit, and Discount shows that discount has a negative relationship with profit, confirming that higher discounts reduce profitability.

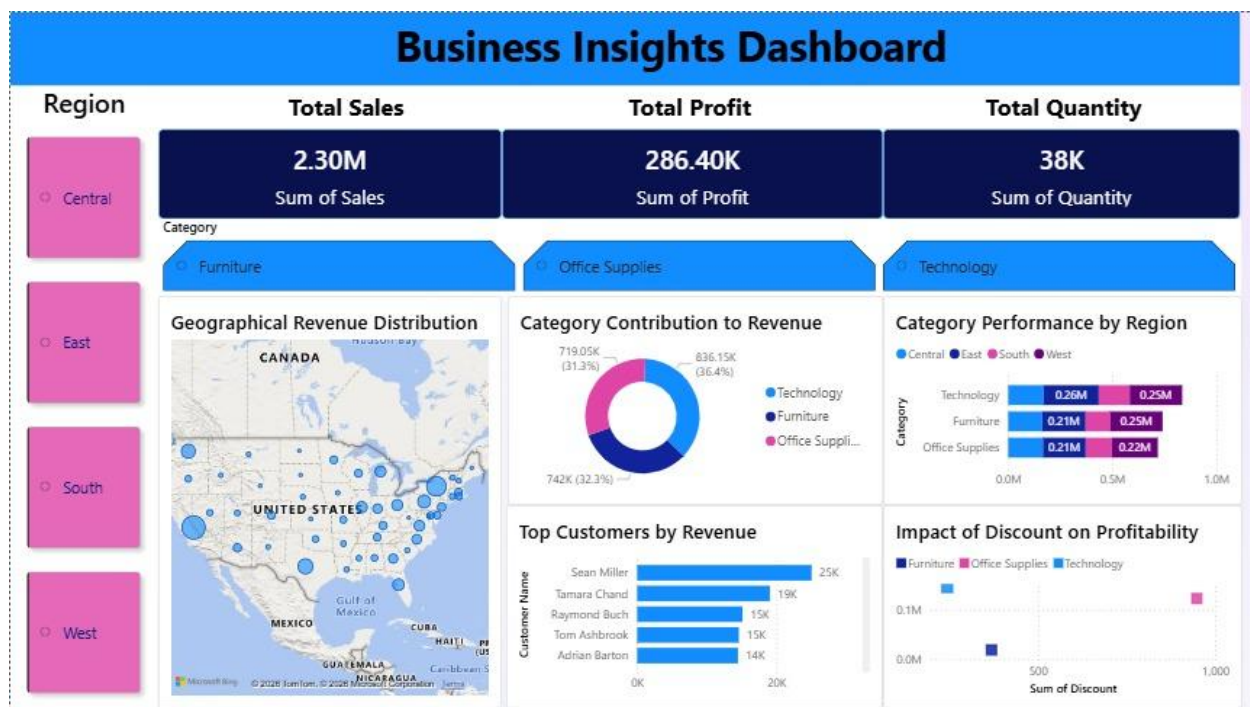
9. Dashboard Development

- Developed an interactive dashboard using Power BI
- Included KPI cards (Sales, Profit, Quantity)
- Added filters for region and category
- Designed layout for clear visualization of insights

Additionally, a React-based dashboard was developed to enhance interactivity and user experience.

✦ *Dashboard screenshots are included below.*

1.Business Insights Dashboard (Power BI)

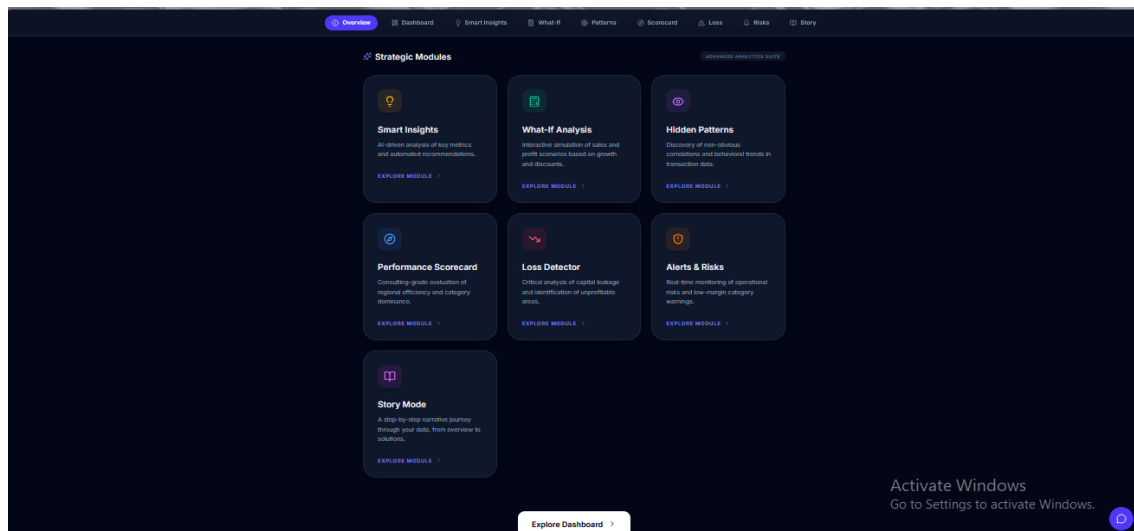
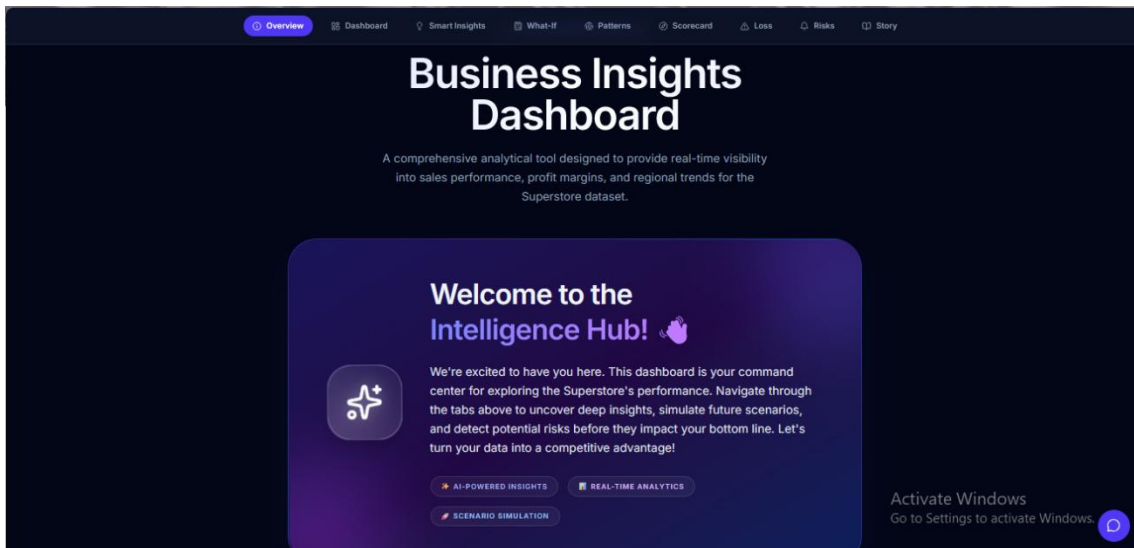


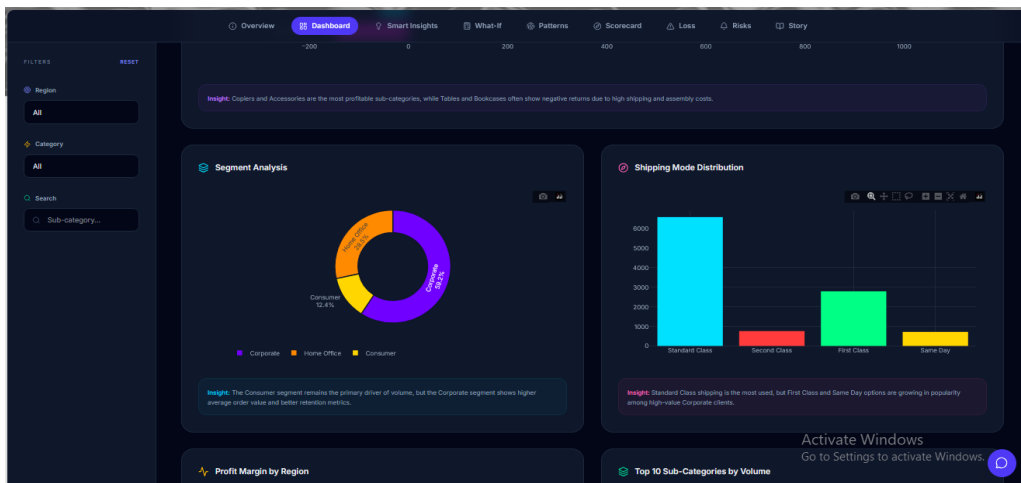
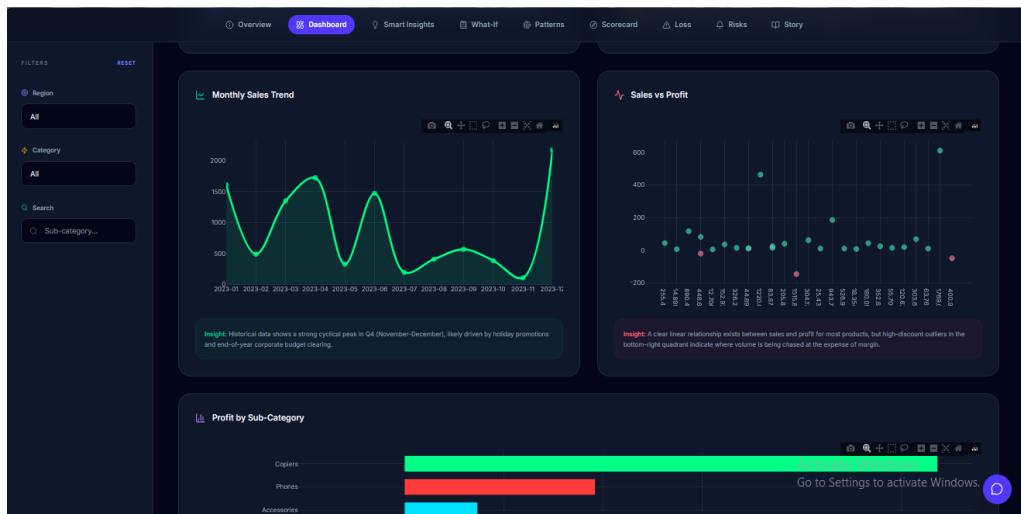
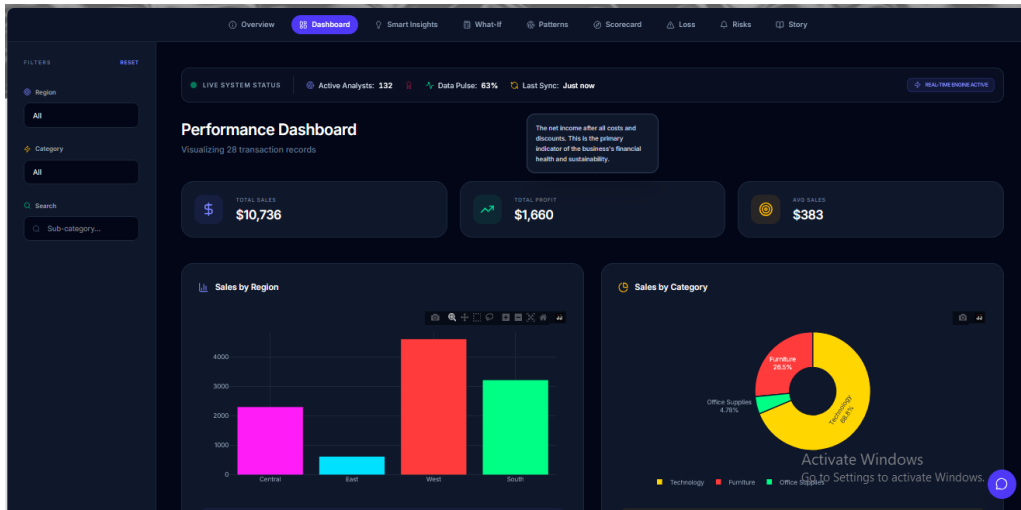
This dashboard provides an interactive overview of business performance using key metrics such as Total Sales, Total Profit, and Total Quantity. It includes multiple visualizations like geographical distribution, category contribution, regional performance, top customers, and the impact of discounts on profitability. Filters for regions allow users to dynamically explore the data.

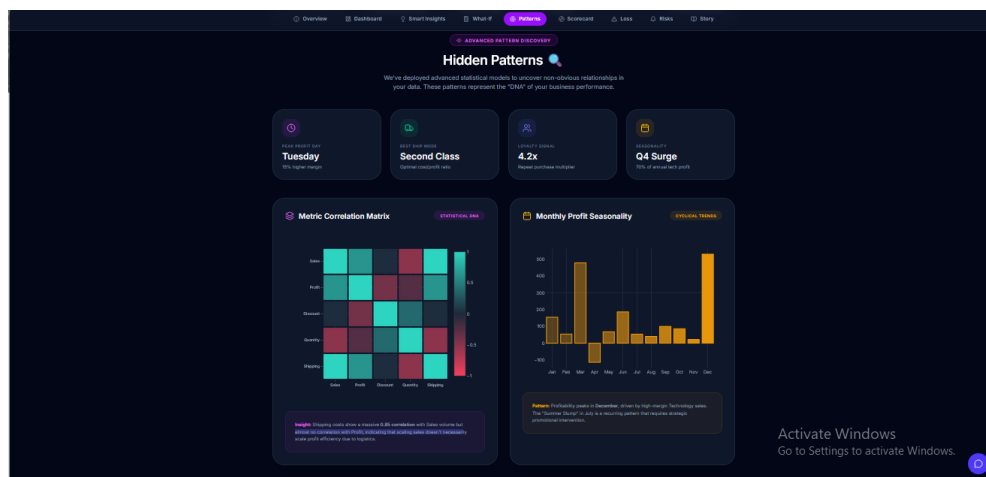
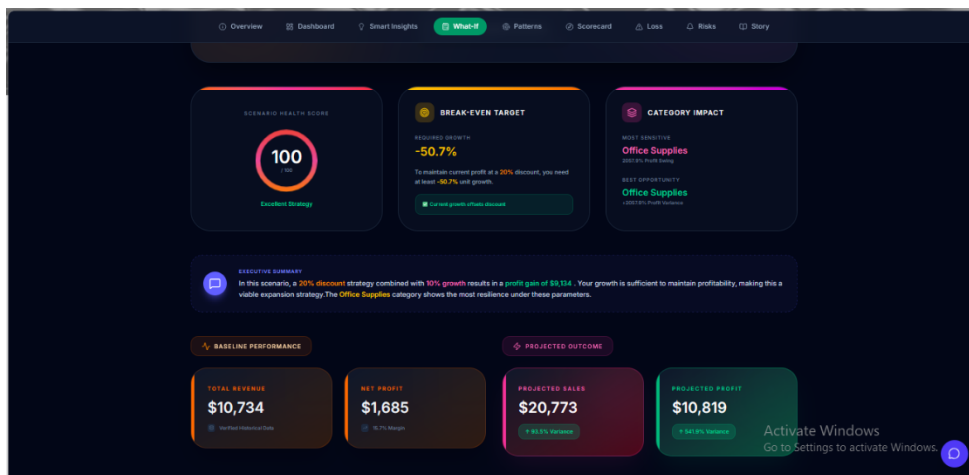
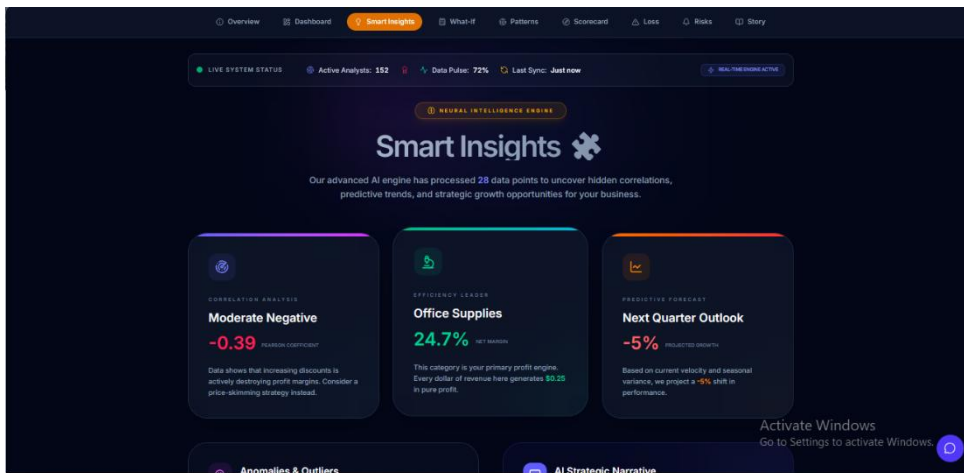
Insight

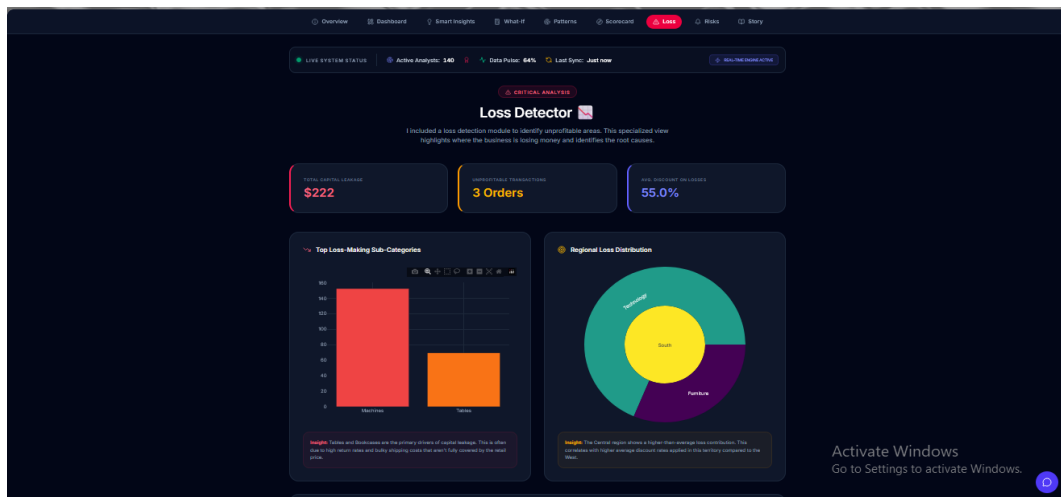
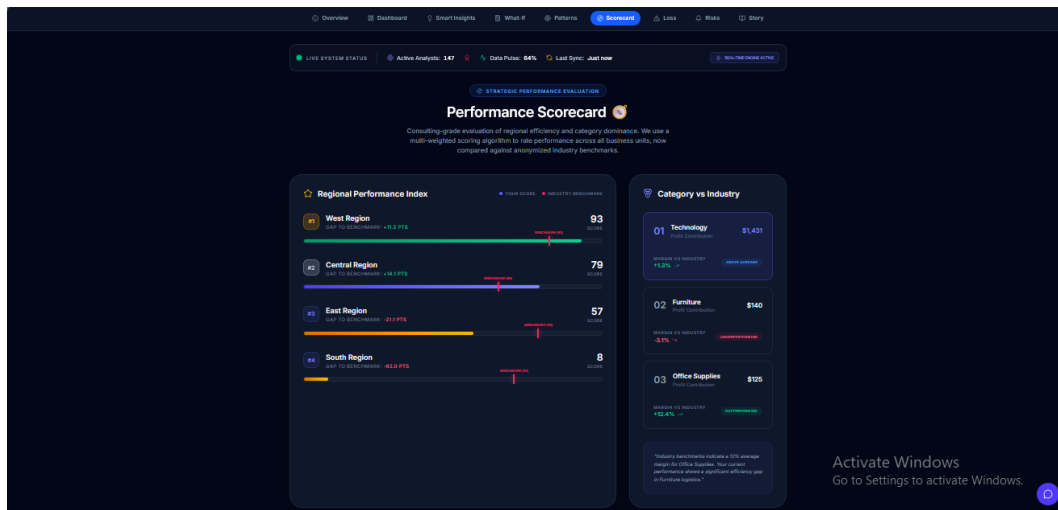
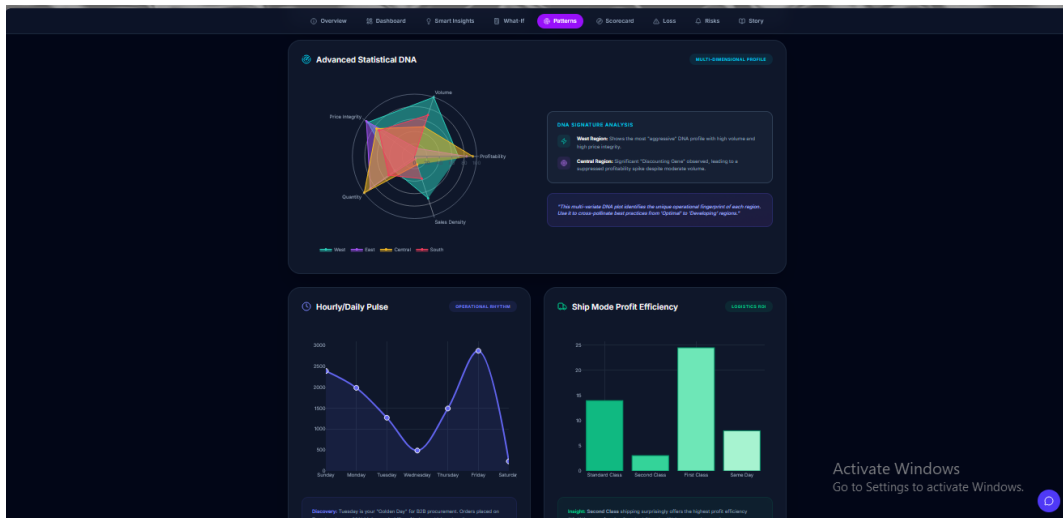
The dashboard highlights that **Technology is the leading category in terms of revenue**, while the **West region contributes significantly to overall sales and profit**. It also shows that higher discounts negatively impact profitability and that a few top customers generate a major portion of revenue.

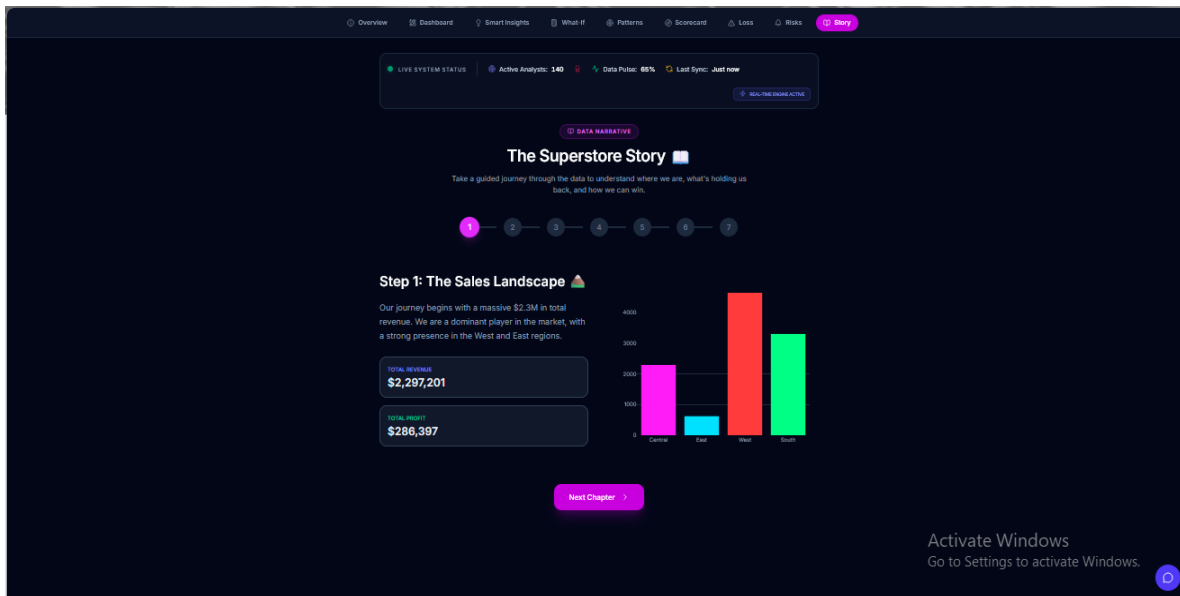
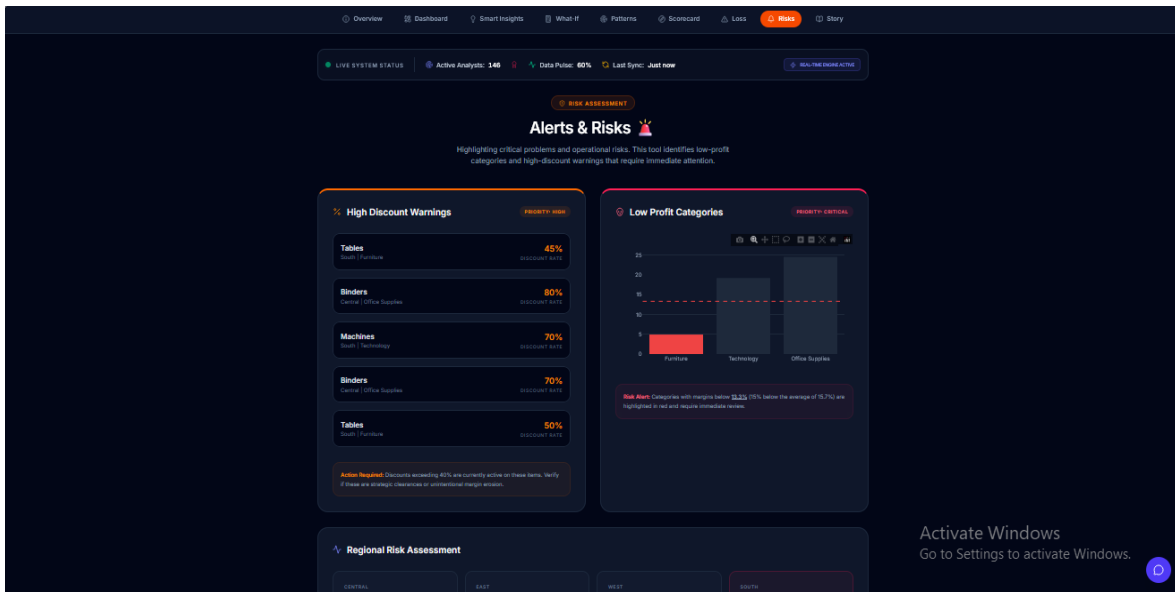
2. Business Insights Dashboard (React Dashboard)











The React dashboard provides an advanced and interactive platform to analyze business data. It is designed to help users understand insights easily and support better decision-making through multiple analytical sections.

a. Overview (Main Page)

This is the starting page of the dashboard. It gives a quick summary of the system and allows users to navigate to different analysis sections. It acts as a central hub for accessing all features.

b. Dashboard (Key Metrics View)

This section shows important business metrics such as Total Sales, Profit, and Average Order Value. It includes charts to analyze performance by region, category, and shipping mode, helping users monitor daily operations.

c. Smart Insights (AI-Based Analysis)

This section provides intelligent insights by analyzing relationships in the data. It highlights important patterns such as how discounts affect profit and identifies high-performing categories. It also explains insights in simple language for better understanding.

d. What-If Analysis (Prediction Tool)

This feature allows users to test different scenarios. For example, users can adjust discount values and see how it affects profit and sales. It helps in planning and decision-making without real risk.

e. Hidden Patterns (Advanced Analysis)

This section shows deeper insights in the data. It helps identify patterns between variables like sales, profit, and quantity, and highlights performance differences across regions.

f. Performance Scorecard

This section evaluates the performance of different regions using key metrics. It provides a score-based comparison to understand which regions are performing better.

g. Loss Detector

This feature helps identify areas where profit is being lost. It highlights situations where high discounts or costs are reducing profitability, allowing businesses to take corrective action.

h. Alerts & Risks

This section shows important warnings such as high discounts or low-profit areas. It helps users quickly identify risks and take action to improve performance.

i. Story Mode (Data Explanation)

This section presents data insights in a simple story format. It helps users understand the journey from sales performance to profit analysis and supports better communication of insights.

10. Project Outputs

- Jupyter Notebook (Data Cleaning, Analysis, Visualizations)
- Power BI Dashboard (.pbix and PDF)
- React Dashboard (Interactive UI)

11. Links

👉 **GitHub Repository (Notebook + Power BI):**

<https://github.com/akshata130504/Business-Insights-Dashboard-using-Superstore-Dataset.git>

👉 **React Dashboard Repository:**

<https://github.com/akshata130504/Business-Insights-Dashboard-using-React-.git>

👉 **React Dashboard Live Deployed Link:**

business-insights-dashboa-git-43abc8-akshata-t-rathods-projects.vercel.app

12. Conclusion

This project successfully transforms raw data into meaningful insights. The analysis and dashboards help in understanding business performance, identifying trends, and supporting better decision-making.